CISS245: Advanced Programming
Quiz q7201

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Q1. Complete the following bubblesort and swap function.
Answer:

```
#include <iostream>

template < typename T >
void swap(T & a, T & b)
{
  T temp = a;
  a = b;
  b = temp;
}

// Sorts values from address start to address end - 1. Note that
// the value at address end is *not* included in the sort. <-- CORRECTION
// Use the above swap function.
template < typename T >
void bubblesort(T * start, T * end)
{
  for(T * i = start; i < end - 1; i++)
  {
    for (T * j = start; j < end - 1 - (i - start); j++)
    {
      if(*j > *(j + 1))
      {
        swap(j, *(j + 1);
      }
    }
  }
}

int main()
{
    int x[] = {5, 3, 1, 2, 4};
    bubblesort< int >(&x[0], &x[4]); // sorts {5, 3, 1, 2} (4 not included)

    double y[] = {7, 5, 3, 1, 2, 4, 6, 8};
    bubblesort< double >(&y[0], &y[6]);

    return 0;
}
```

Q2. Complete the following `Ptr` class which is similar to our `IntPtr` class except that `Ptr` is a class template. The following test code should work with your `Ptr` class. Of course there should be no memory issues.

```
#include <iostream>
#include "Ptr.h"

int main()
{
    Ptr< int > p(42);
    std::cout << *p << '\n';

    Ptr< int > q(p);
    std::cout << *q << '\n';

    Ptr< int > r(0);
    r = p;
    std::cout << *r << '\n';

    // Similar test cases for Ptr< double >, Ptr< char >, etc. should work.
    return 0;
}
```

Answer:

```
# file: Ptr.h

#ifndef PTR_H
#define PTR_H

#include <iostream>

template < typename T >
class Ptr
{
public:
    Ptr(const T & val)

    {
      p_ = new T(val)
    }
    ~Ptr()
    {
      delete p_;
    }
    Ptr(const Ptr< T > & p)

    {
      p_ = new T(*other.p_);
    }
    const Ptr< T > & operator=(const Ptr< T > & p)
```

```
    {
      if (this != &p)
      {
        delete p_;
        p_ = new T (*p.p_);
      }
      return *this;
    }
    T & operator*()
    {
      return *p_;
    }
private:
    T * p_;
};

// you need something here
template < typename T >
std::ostream & operator<<(std::ostream & cout, const Ptr< T > & p)
{
  return cout << *p;
}

#endif
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.